

STAT 482 — Case Study 5

Modeling Environmental Factors on Ecosystem Stress

Multiple Linear Regression — Forest Fires Dataset (UCI)

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What's the Data?

Forest Fires dataset — 517 fire events from Montesinho Park, Portugal

517

Fire events

8

Predictors

0

Missing values

12.85

Mean area (ha)

Predictors (Independent)

- FPMC — Fine Fuel Moisture Code
- DMC — Duff Moisture Code
- DC — Drought Code
- ISI — Initial Spread Index
- temp — Temperature (°C)
- RH — Relative Humidity (%)
- wind — Wind Speed (km/h)
- rain — Rainfall (mm)

Response (Dependent)

Burned Area (ha)

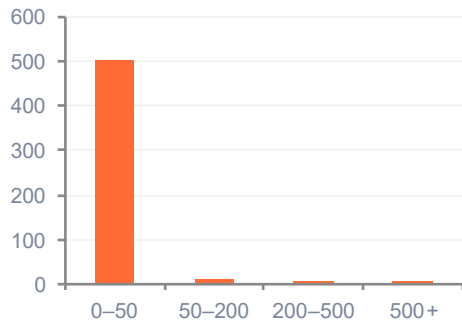
Heavily right-skewed — most fires burned 0 ha, a few burned 1,000+ ha.

Fix: We use $\log(\text{area} + 1)$ to make the distribution usable for regression.

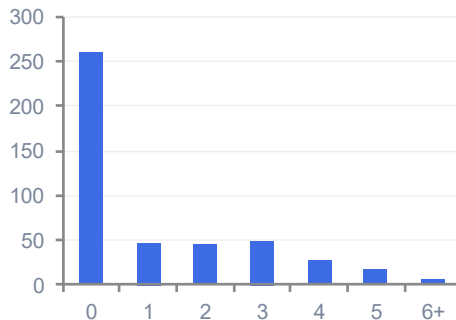
Exploratory Data Analysis

Distributions, transformation, and correlations between variables

Original area — very skewed



$\log(\text{area}+1)$ — much better!



The log transformation compresses extreme values and makes the distribution closer to normal — a key OLS requirement.

Correlations with $\log_area$

Variable	r	Notable pair
DMC	0.07	DMC-DC: 0.68
wind	0.07	—
DC	0.07	DC-DMC: 0.68
temp	0.05	temp-RH: -0.53
RH	-0.05	RH-temp: -0.53
rain	0.02	—

Key takeaway

No single variable is strongly correlated with the response — burned area is tough to predict from weather alone. Highest r is only 0.07.

Full Regression Model — Results

8 predictors, OLS on log(area+1), N = 517

$$\log(\text{area}+1) = 0.222 + 0.008 \cdot \text{FFMC} + 0.001 \cdot \text{DMC} + 0.0003 \cdot \text{DC} - 0.024 \cdot \text{ISI} + 0.002 \cdot \text{temp} - 0.005 \cdot \text{RH} + 0.076 \cdot \text{wind}^* + 0.097 \cdot \text{rain}$$

Predictor	Estimate	Std. Error	t-value	p-value
(Intercept)	0.2224	1.3604	0.163	0.870
FFMC	0.0077	0.0145	0.532	0.595
DMC	0.0012	0.0015	0.814	0.416
DC	0.0003	0.0004	0.767	0.444
ISI	-0.0239	0.0169	-1.415	0.158
temp	0.0025	0.0173	0.143	0.887
RH	-0.0052	0.0052	-0.997	0.319
wind 	0.0758	0.0366	2.069	0.039
rain	0.0965	0.2121	0.455	0.649

Model Fit

R ²	0.020
Adj. R ²	0.004
F-stat	1.288
F p-value	0.247
Residual SE	1.395

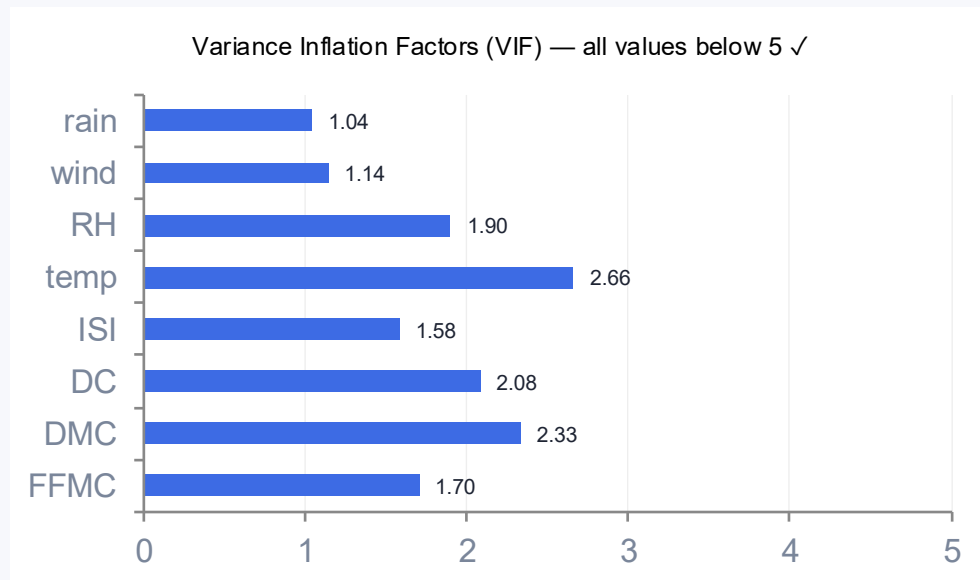
What does this mean?

Only wind is significant ($p = 0.039$). The model explains less than 1% of variance — environmental variables alone aren't enough to predict fire size.

 = statistically significant at $p < 0.05$

Checking Model Assumptions

— Multicollinearity (VIF), normality, and homoscedasticity tests



Rule of thumb: VIF < 5 is fine, > 10 is a serious problem

✓ PASS

Breusch-Pagan

BP = 11.27, $p = 0.187$ · No heteroscedasticity problem

✗ FAIL

Shapiro-Wilk

W = 0.854, $p < 2.2e-16$ · Residuals are non-normal (heavy tail)

✓ PASS

VIF (highest)

temp = 2.66 · All predictors below threshold

⚠ WARN

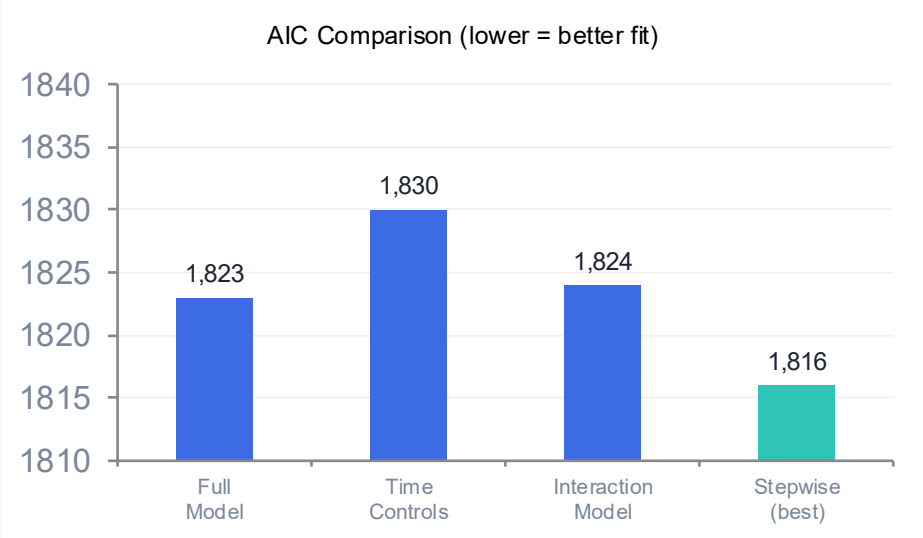
Residuals vs Fitted

Fan-shaped pattern · Slight heteroscedastic tendency visible

Non-normality of residuals is expected here — the log transform helps but extreme fires still affect the tail. With $N=517$ the CLT protects our inference.

Which Model Is Best?

Comparing full model, time controls, interaction term, and stepwise selection



BEST MODEL

Stepwise Result (stepAIC)

$$\log(\text{area}+1) = 0.913 + 0.00175 \cdot \text{DMC} - 0.00558 \cdot \text{RH} + 0.0624 \cdot \text{wind}$$

Predictor	β	p-value
(Intercept)	0.913	<0.001 ***
DMC	0.00175	0.070 .
RH	-0.00558	0.140
wind	0.0624	0.071 .

All Models Compared				
Model	Adj. R ²	AIC	BIC	Verdict
Full (8 pred.)	0.004	1823	1865	Weak
Time Controls	0.021	1830	1945	Moderate
Interaction	0.004	1824	1871	Weak
★ Stepwise	0.008	1816	1837	Best AIC/BIC

How Well Does It Predict?

Train/Test split (80/20) using the caret package

Data Split

Train: 416 obs (80%)

Test: 101 obs (20%)

set.seed(202031137)

1.71

RMSE (test set) — log scale

1.30

MAE (test set) — log scale

Training Model Coefficients

Predictor	β	p-value
FFMC	0.00455	0.759
DMC	0.00050	0.755
DC	0.00060	0.110
ISI	-0.02989	0.088
temp	0.00623	0.739
RH	-0.00298	0.597
wind ★	0.08286	0.037 *
rain	-0.89116	0.237

What do these numbers mean?

- RMSE = 1.71 on log scale → roughly ± 5.5 ha error on original scale.
- MAE = 1.30 → average error ≈ 3.7 ha. Big uncertainty for large fires.
- Wind is still the only significant predictor in training too ($p = 0.037$).
- Low R^2 ($\approx 2\%$) confirms the model is weak — weather data alone can't fully explain fire size.
- This is a well-known challenge in fire science literature.

Conclusions & Takeaways

1

Wind is the #1 predictor

Only variable significant across all models ($\beta = 0.076$, $p = 0.039$).
Higher wind $\rightarrow$ bigger fires.

2

Stepwise gives the cleanest model

DMC + RH + wind retained. Lowest AIC (1815) and BIC (1837) of all candidates.

3

No multicollinearity issues

All VIF values < 5 . The weak R^2 is a real data problem, not a statistics artifact.

4

Formal tests mostly pass

Homoscedasticity holds (BP $p = 0.187$). Normality fails (SW $p < 2.2e-16$) — expected with fire data.

5

Model has real limitations

RMSE = 1.71, MAE = 1.30 on log scale. Need spatial/topographic data to predict fire size well.